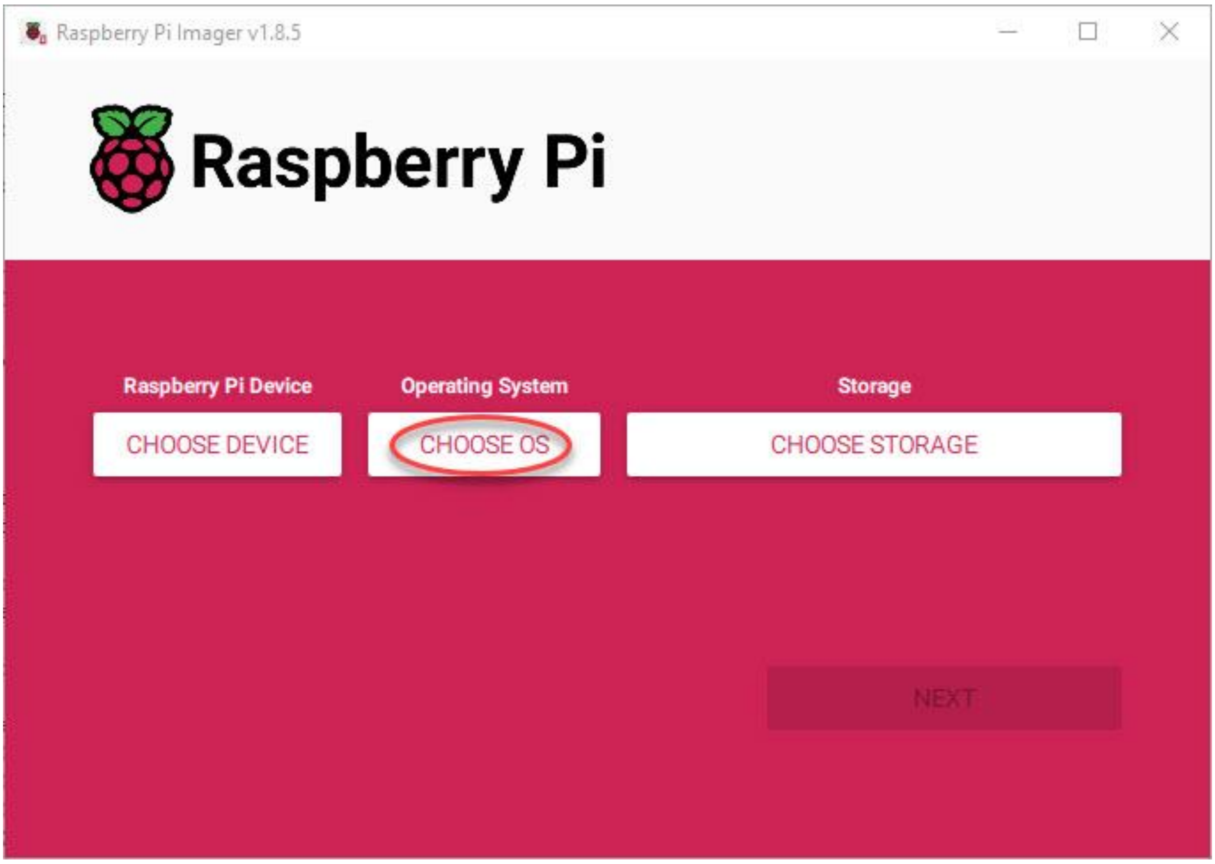
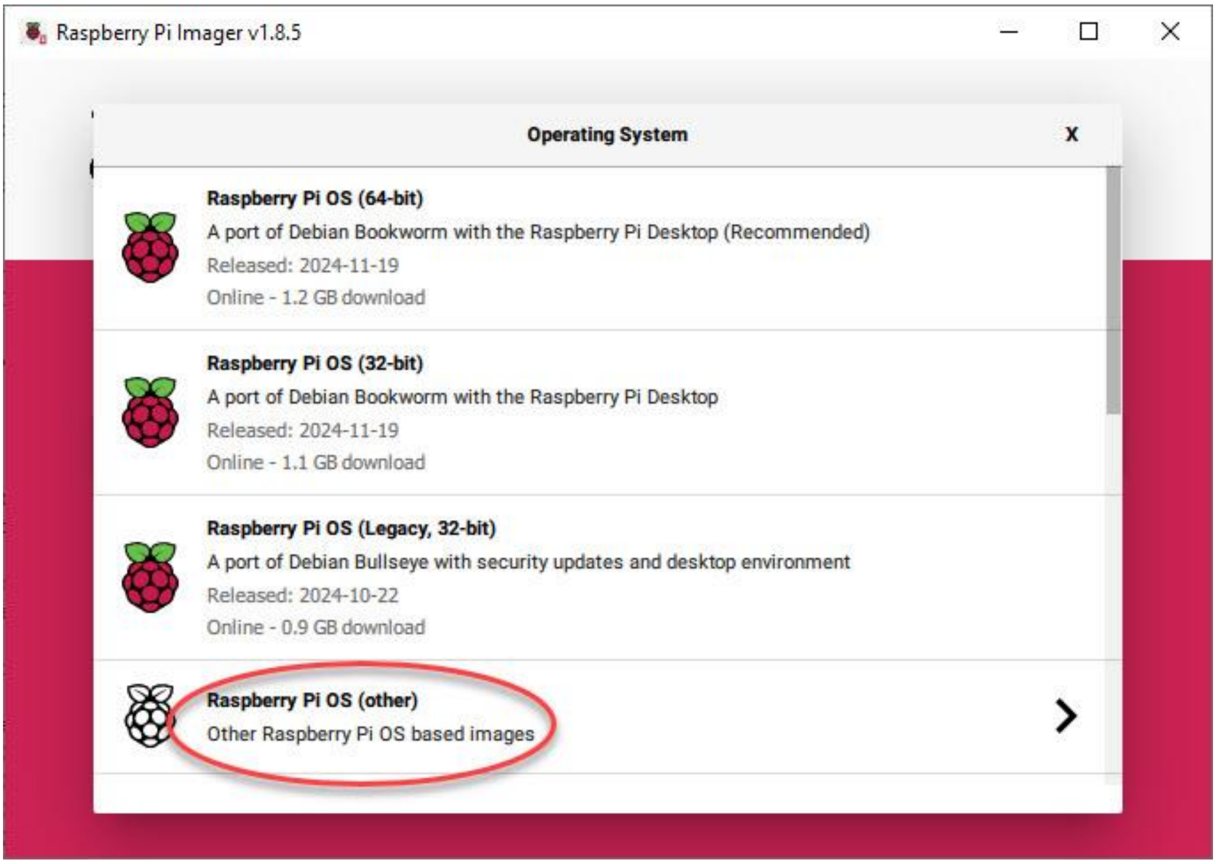
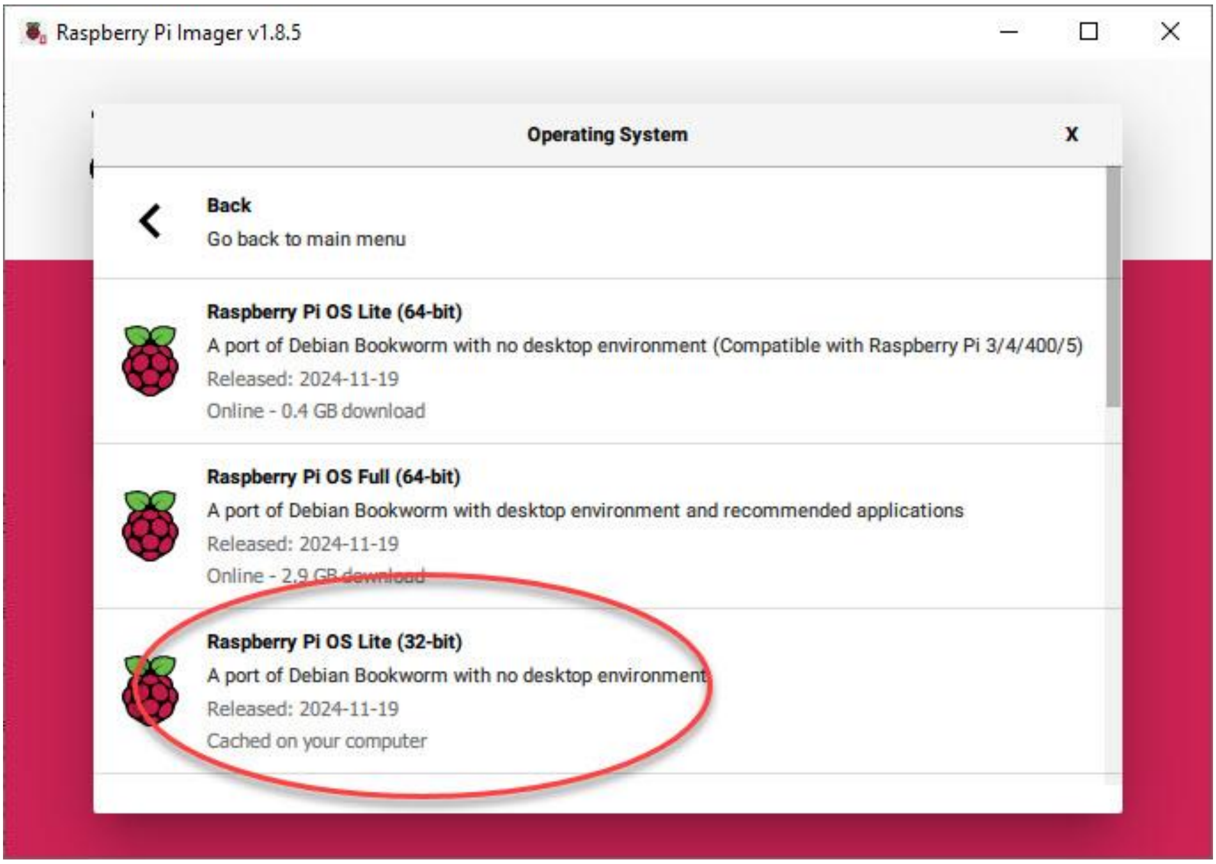
Pi-Ager install on Raspberry Pi 3/Pi 4/Pi 5/Pi zero (2)w under Pi OS 12 Lite bookworm

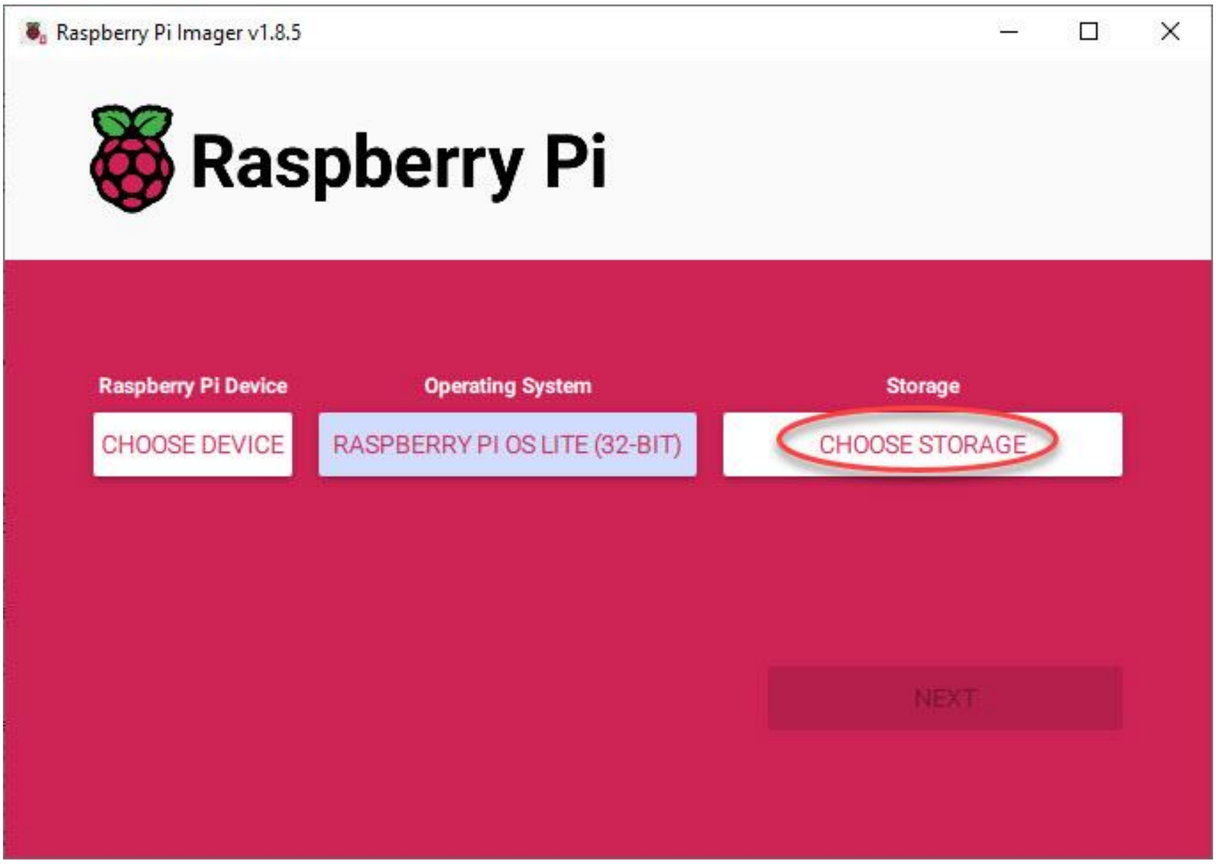
- For all Pi:
  Download and install Raspberry Pi Imager v1.8.4 or later from
  https://www.raspberrypi.com/software/.

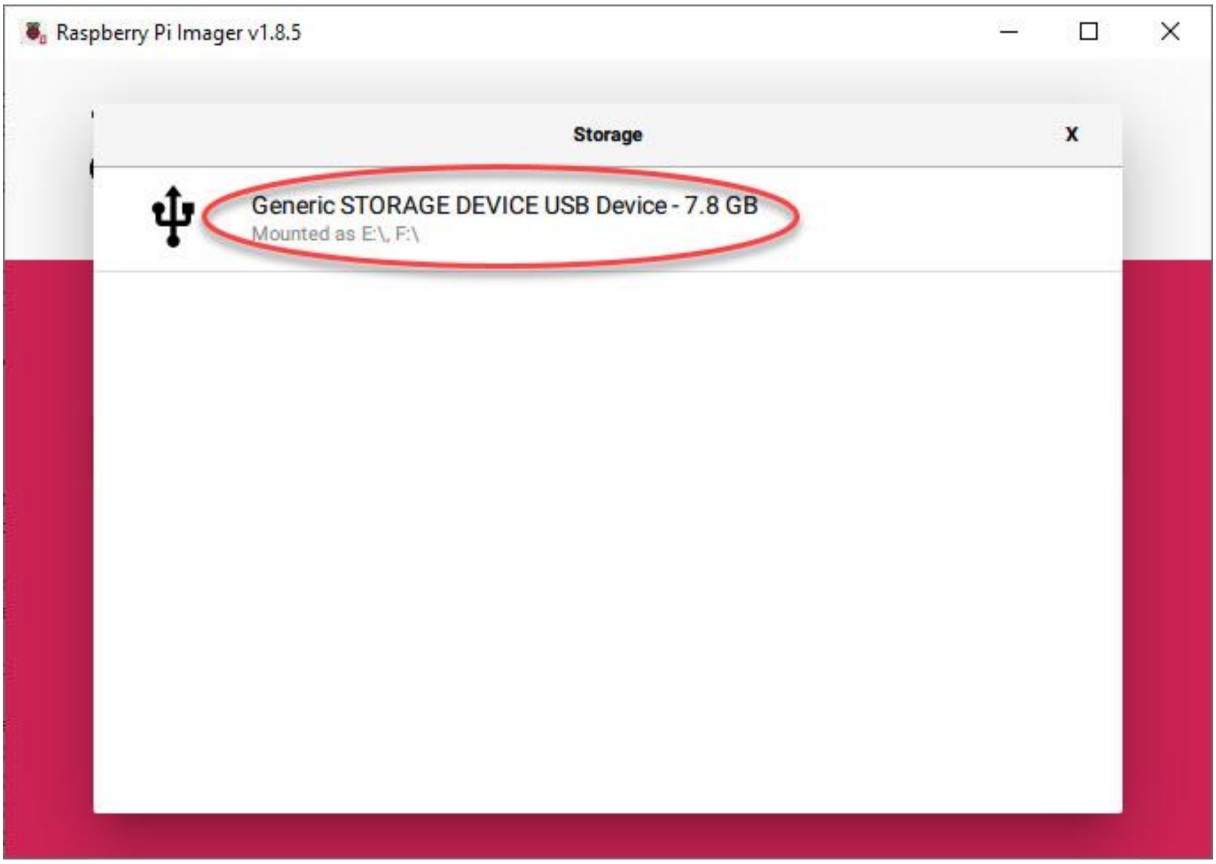
  Start Raspberry Pi Imager, then continue with the follow steps:

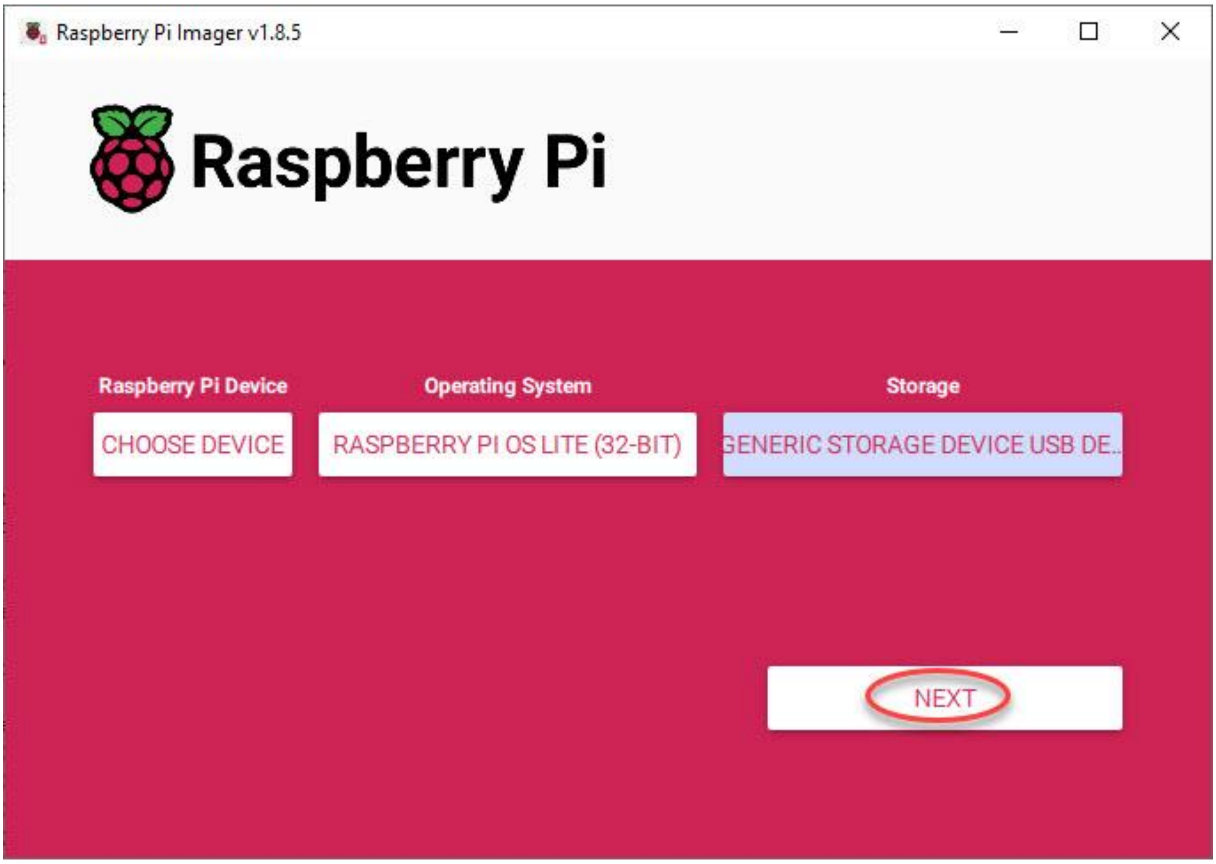


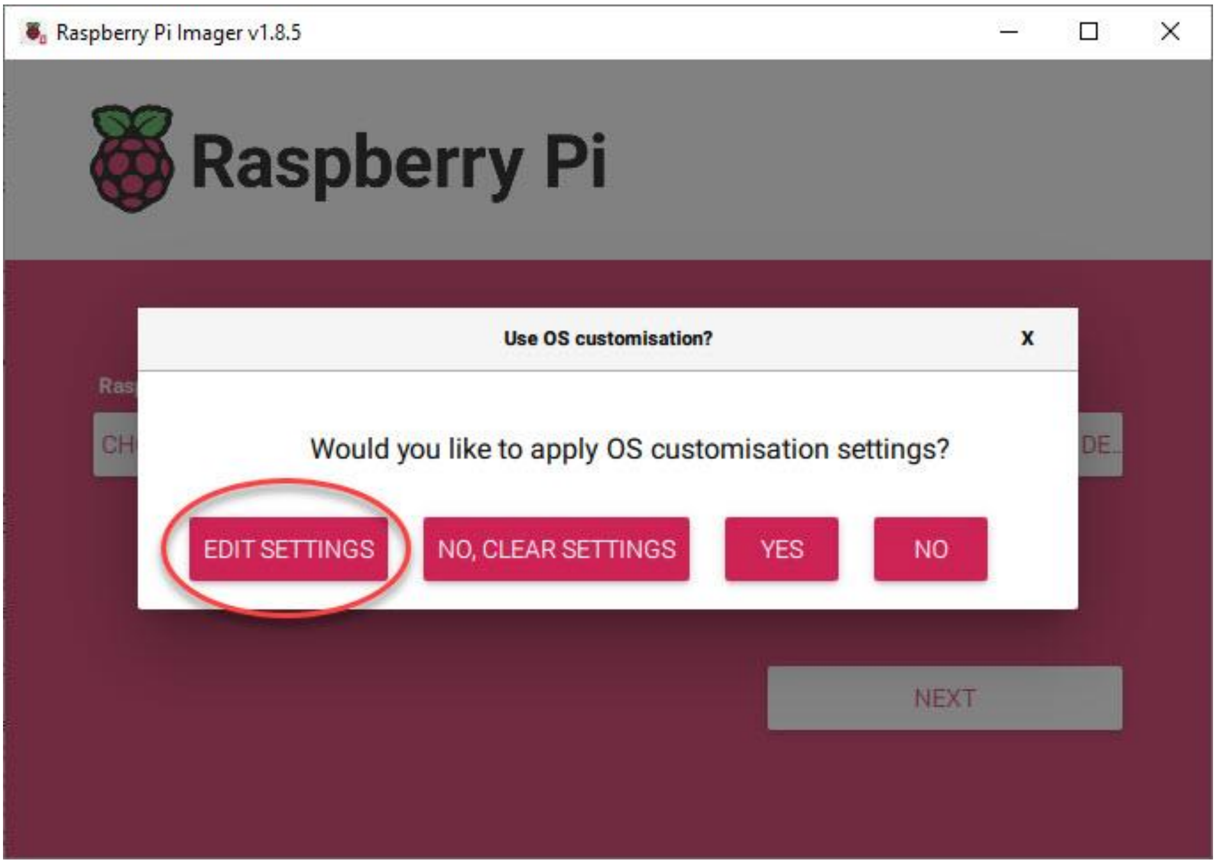

Raspberry Pi Imager v1.8.5
Operating System
X
Raspberry Pi OS (64-bit)
A port of Debian Bookworm with the Raspberry Pi Desktop (Recommended)
Released: 2024-11-19
Online - 1.2 GB download
Raspberry Pi OS (32-bit)
A port of Debian Bookworm with the Raspberry Pi Desktop
Released: 2024-11-19
Online - 1.1 GB download
Raspberry Pi OS (Legacy, 32-bit)
A port of Debian Bullseye with security updates and desktop environment
Released: 2024-10-22
Online - 0.9 GB download
Raspberry Pi OS (other)
Other Raspberry Pi OS based images

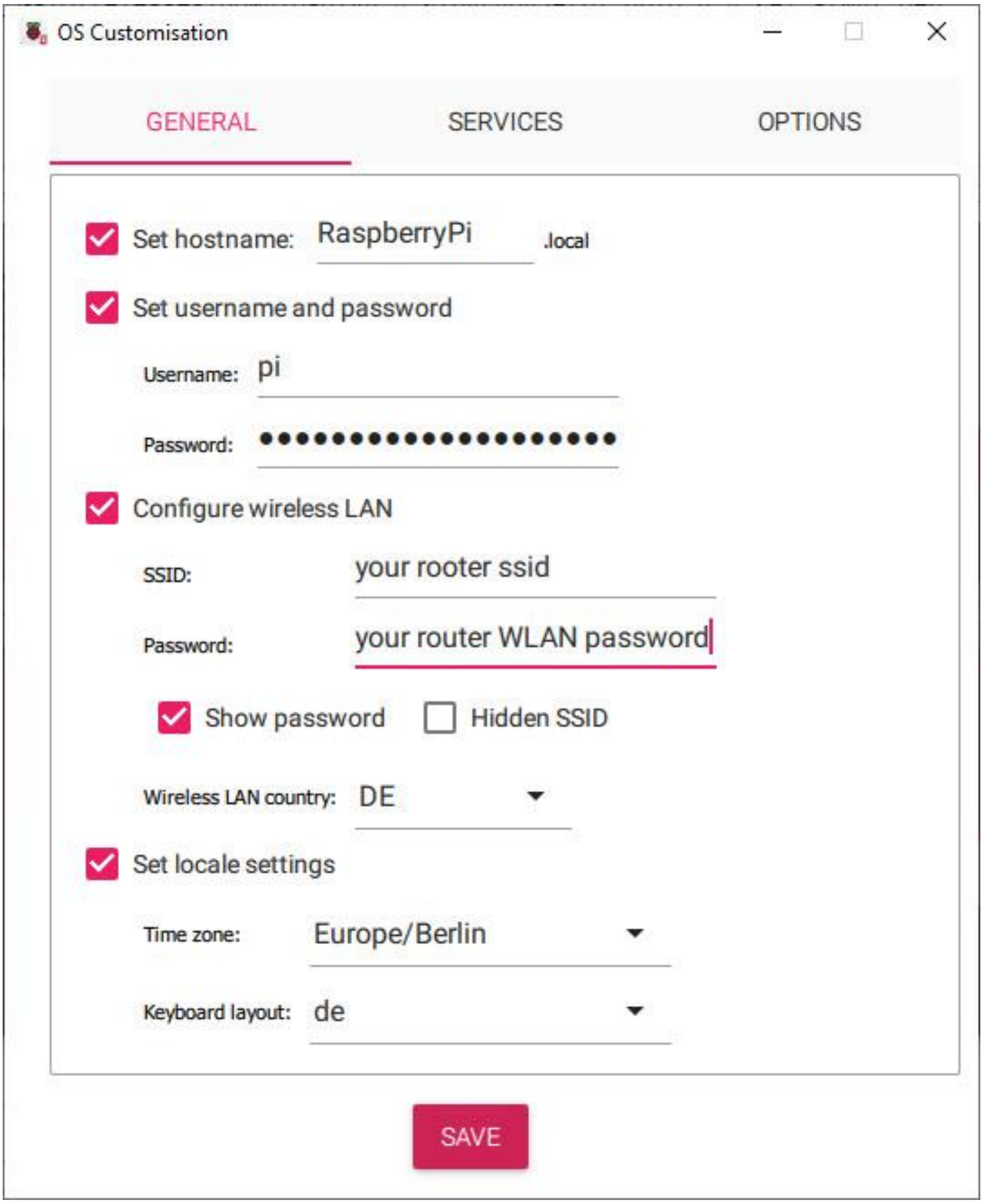

Raspberry Pi Imager v1.8.5
Operating System
X
Back
Go back to main menu
Raspberry Pi OS Lite (64-bit)
A port of Debian Bookworm with no desktop environment (Compatible with Raspberry Pi 3/4/400/5)
Released: 2024-11-19
Online - 0.4 GB download
Raspberry Pi OS Full (64-bit)
A port of Debian Bookworm with desktop environment and recommended applications
Released: 2024-11-19
Online - 2.9 GB download
Raspberry Pi OS Lite (32-bit)
A port of Debian Bookworm with no desktop environment
Released: 2024-11-19
Cached on your computer

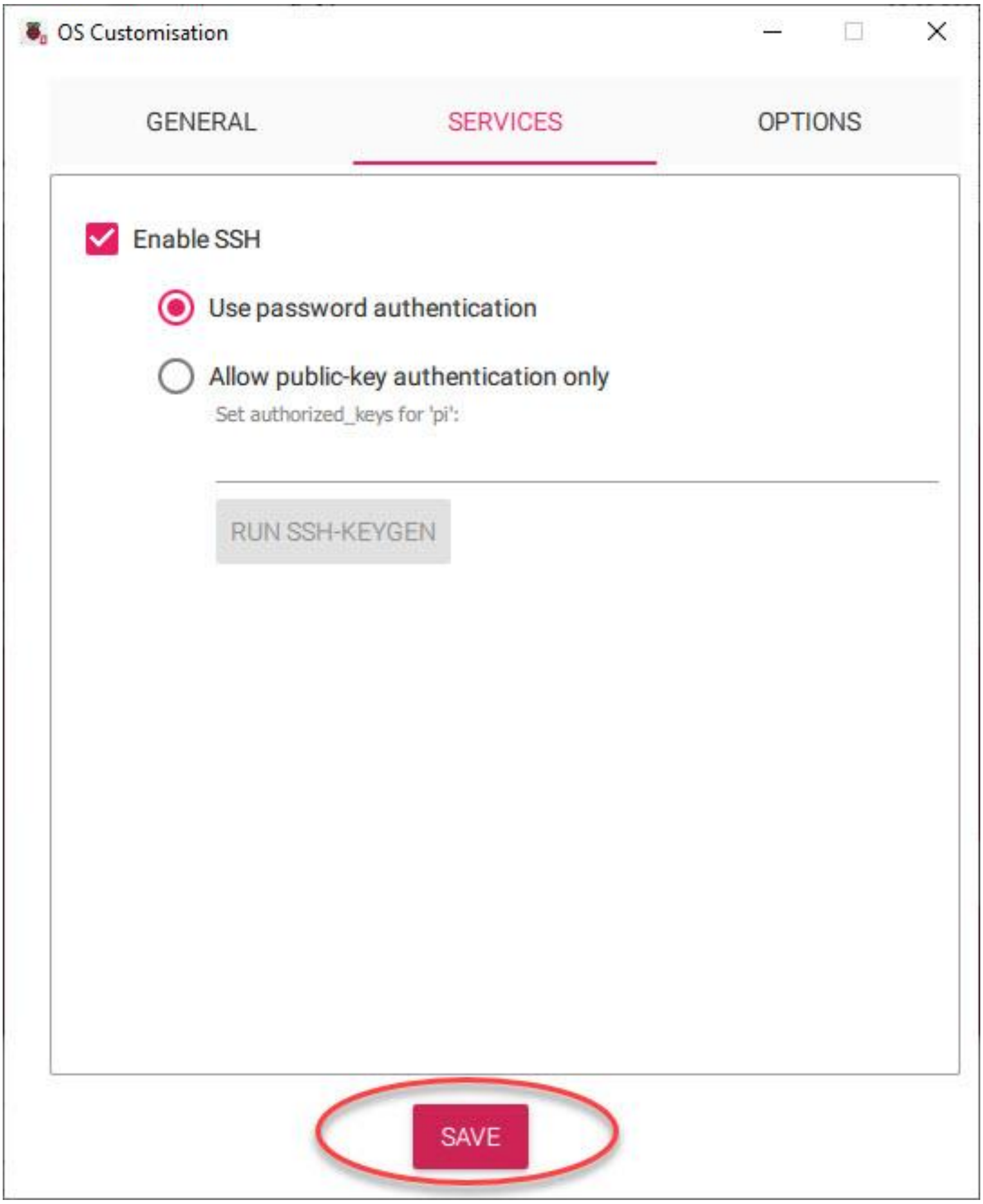

Raspberry Pi Imager v1.8.5
Raspberry Pi
Raspberry Pi Device
Operating System
Storage
CHOOSE DEVICE
RASPBERRY PI OS LITE (32-BIT)
CHOOSE STORAGE
NEXT

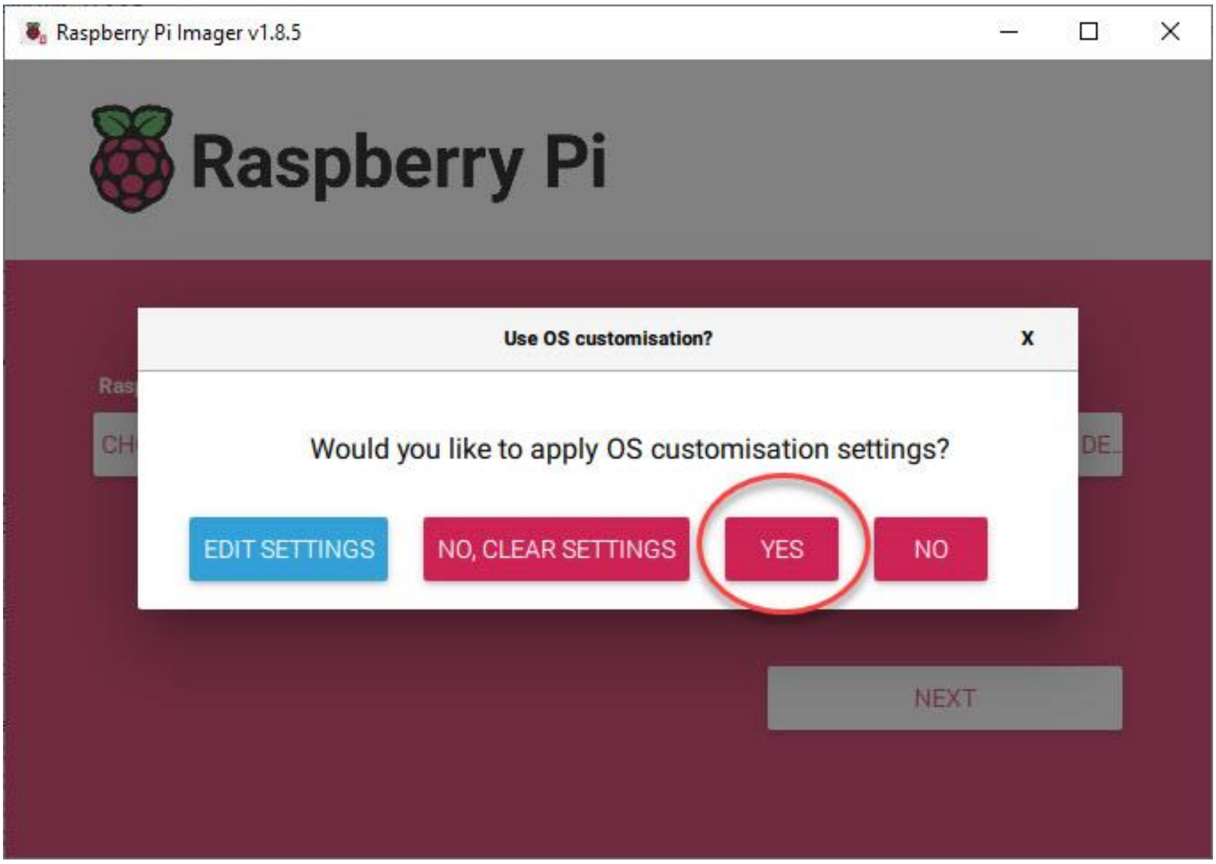

Raspberry Pi Imager v1.8.5
Storage
X
Generic STORAGE DEVICE USB Device - 7.8 GB
Mounted as E:\, F:\

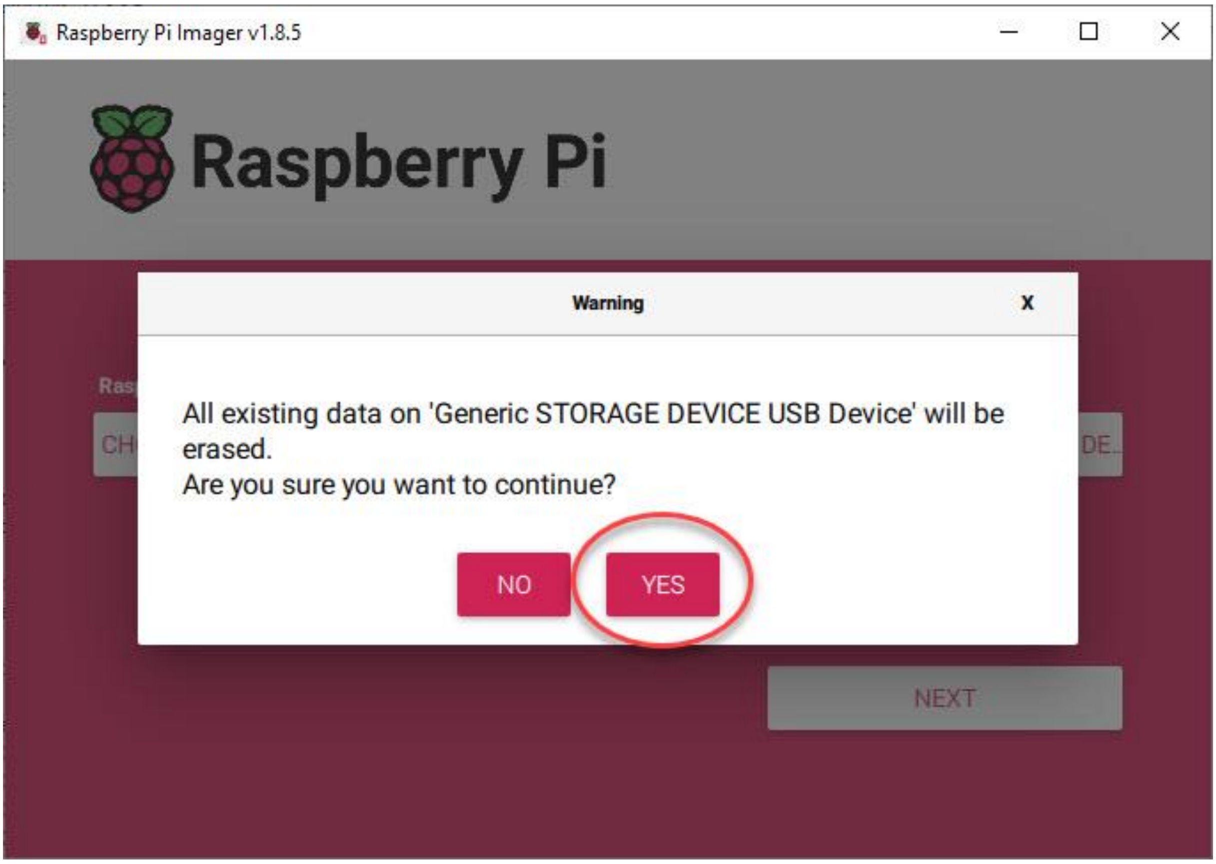

Raspberry Pi Imager v1.8.5
Raspberry Pi
Raspberry Pi Device
CHOOSE DEVICE
Operating System
RASPBERRY PI OS LITE (32-BIT)
Storage
GENERIC STORAGE DEVICE USB DE...
NEXT


Raspberry Pi Imager v1.8.5
Raspberry Pi
Use OS customisation?
X
Would you like to apply OS customisation settings?
EDIT SETTINGS
NO, CLEAR SETTINGS
YES
NO
NEXT


OS Customisation
GENERAL
SERVICES
OPTIONS
Set hostname: RaspberryPi .local
Set username and password
Username: pi
Password: ••••••••••••••••••••
Configure wireless LAN
SSID: your rooter ssid
Password: your router WLAN password
Show password
Hidden SSID
Wireless LAN country: DE
Set locale settings
Time zone: Europe/Berlin
Keyboard layout: de
SAVE


OS Customisation
GENERAL
SERVICES
OPTIONS
Enable SSH
Use password authentication
Allow public-key authentication only
Set authorized_keys for 'pi':
RUN SSH-KEYGEN
SAVE





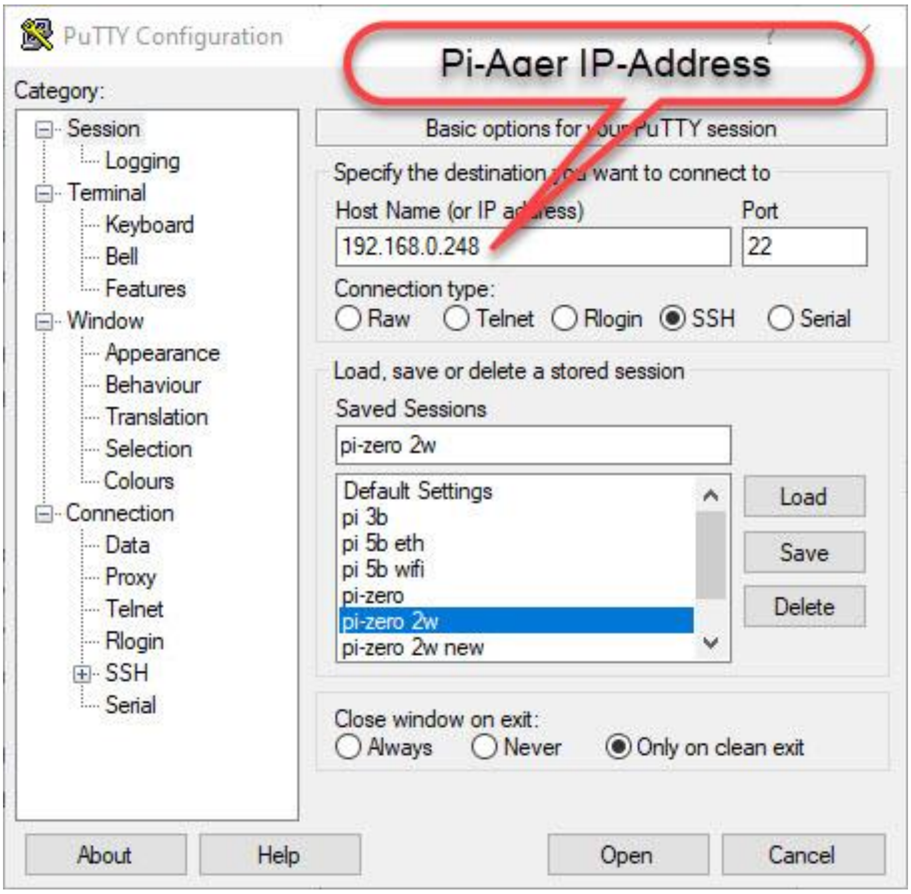
Then ‘SAVE’ and apply OS customization settings with ‘YES’.
The customized OS is now written to your SD-Card.

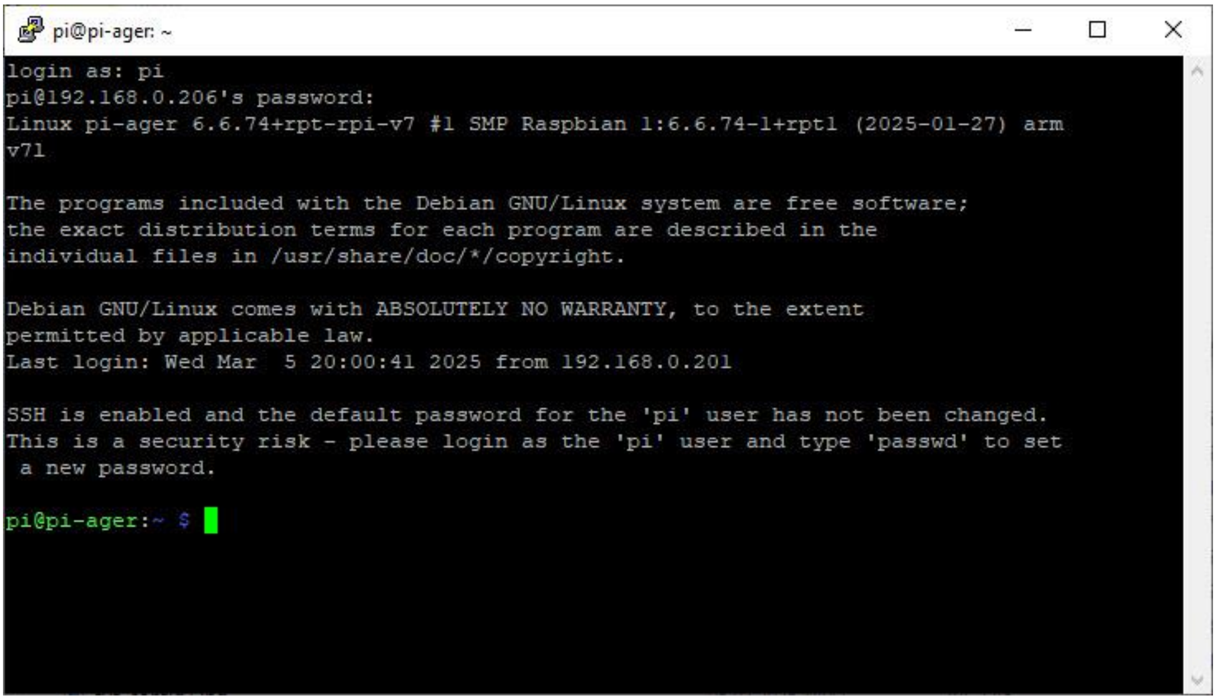
Put your SD-Card in your Raspberry Pi and power on your Pi device.
Login via SSH (e.g. use PuTTY) or connect a HDMI monitor and USB keyboard to your Raspberry device and continue with the setup as described below.

- Download and install PuTTY from https://www.putty.org/

- Optional download and install WinSCP from https://winscp.net/eng/index.php

- Start PuTTY to connect to Pi-Ager:





- Download system build scripts from github:
  wget -O pi-ager-build-system.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-build-system.sh

  wget -O pi-ager-finalize-build.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-finalize-build.sh

  chmod +x *.sh

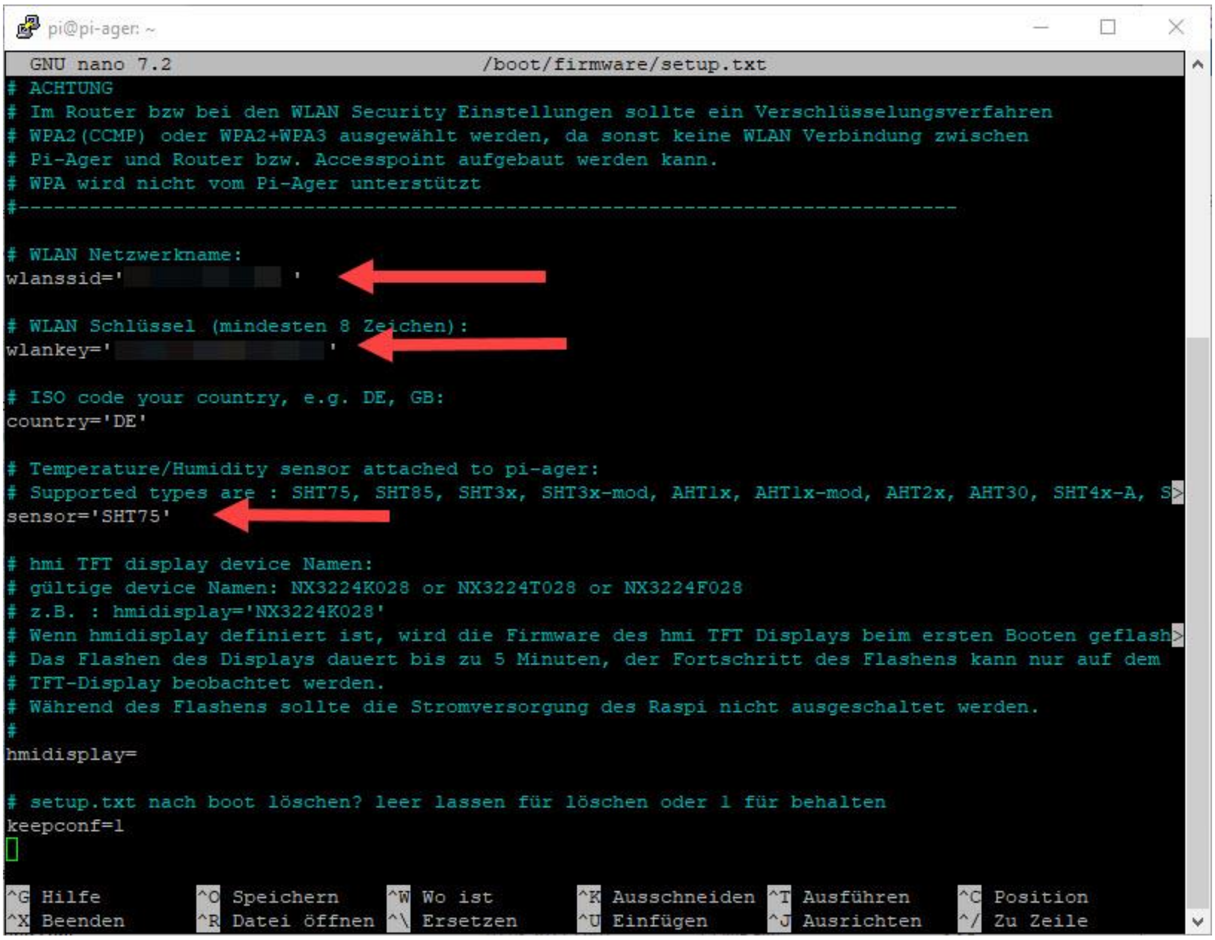
  Start now first phase of Pi-Ager System setup:
  sudo ./pi-ager-build-system.sh

  You can now follow the system setup, that is divided in 3 steps with rebooting between the steps. Restart the script after rebooting to continue the system setup :

  sudo ./pi-ager-build-system.sh

  After step 2 you are informed to edit the file /boot/firmware/setup.txt that contains information about passwords, WLAN setup parameter, temperature/humidity sensor to use and optional HMI display version connected to your Pi-Ager:

  sudo nano /boot/firmware/setup.txt



After saving setup.txt the last step initializes the Pi-Ager system with data from setup.txt by starting the script pi-ager-finalize-build.sh:

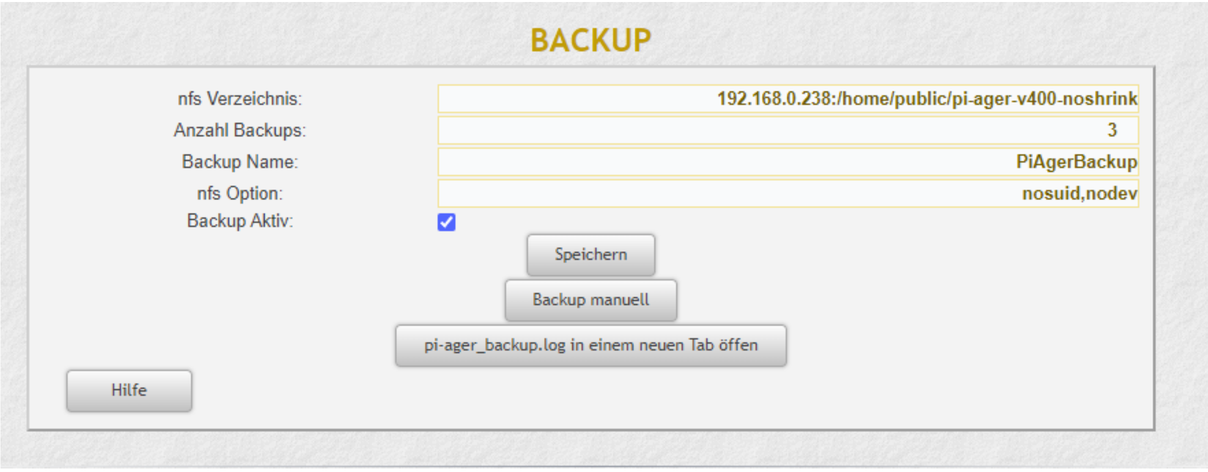
sudo ./pi-ager-finalize-build.sh

The system is then rebooted and ready to start the Pi-Ager. Open your browser window and enter the IP-Address of the Pi-Ager in the address line.

How to generate and deploy Pi-Ager Images

There are some steps to do for generating and deploying a new image from a running Pi-Ager system:

1. Setup a NFS Server to store a new image. The best way is using a Raspi 4B or 5 equipped with an USB Memory stick with 128GB or more. Using RPi 5 it is now possible to add a NVME SSD so that there will be enough space for storing one or more images.
   How to setup a NFS Server :
   https://www.elektronik-kompendium.de/sites/raspberry-pi/2007061.htm
2. On the Pi-Ager setup and start an Image backup on the ADMIN page. For example :



3. Next start a terminal on the Pi-Ager and start the pi-ager_image.sh script with option -l: (lowercase ‘L’)

```
pi@pi-ager:~ $ cd /usr/local/bin/
pi@pi-ager:/usr/local/bin $ sudo pi-ager_image.sh -l
Backup ist inaktiv. Pi-Ager_image wird gestartet!
überprüfe ob der NFS-Server vorhanden ist.
Checking...
192.168.0.238:/home/public/pi-ager-v400-noshrink ist vorhanden
überprüfe ob Backup Pfad vorhanden ist.
Checking ...
/home/nfs/public ist vorhanden
überprüfe ob NFS Mount point definiert ist.
Checking ...
/home/nfs/public ist definiert
überprüfe ob NFS Mount point im file system angelegt ist.
Checking...
/home/nfs/public ist angelegt
NFSVOL=192.168.0.238:/home/public/pi-ager-v400-noshrink
NFSPATH=/home/nfs/public
NFSMOUNT=/home/nfs/public
umount: /home/nfs/public: not mounted.
hänge NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink ein
mount with options: nosuid,nodev
mount NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink successfull. Image script continues
.
Load last backup file.
Backup path is  /home/nfs/public
Backup file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Backup path with file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Source File = /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
do_copy     = false
my_image    = false
Using /home/nfs/public/PiAgerBackup_2024-02-12-103529.img as source and target
#####################################################################################
parted_output = 2:541065216B:15931539455B:15390474240B:ext4::;
partnum = 2
partstart = 541065216
loopback = /dev/loop0
#####################################################################################
parted_output_boot = 1:4194304B:541065215B:536870912B:fat32::lba;
partnum_boot = 1
partstart_boot = 4194304
loopback_boot = /dev/loop1
#####################################################################################
mount directory is /tmp/tmp.SfM3akt9CW
2024-02-12 10:39:10 URL:https://raw.githubusercontent.com/Tronje-the-Falconer/Pi-Ager/entwicklung/boo
t/firmware/setup.txt [3281/3281] -> "setup.txt" [1]
setup.txt copied to /tmp/tmp.SfM3akt9CW/boot/
cmdline.txt modified, added init=/usr/lib/raspberrypi-sys-mods/firstboot
rematch1 /tmp/
rematch2 tmp.SfM3akt9CW
change rootdir is tmp.SfM3akt9CW
/tmp
pi-ager_main.service disabled, will be enabled during setup_pi-ager.sh
Created symlink /etc/systemd/system/multi-user.target.wants/setup_pi-ager.service → /lib/systemd/system/setup_p
i-ager.service.
setup_pi-ager service enabled
Running in chroot, ignoring command 'is-active'
hostname changed to pi-ager
unmount /tmp/tmp.SfM3akt9CW/boot
unmount /tmp/tmp.SfM3akt9CW
Detaching loop devices from /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
The image /home/nfs/public/PiAger_image_2024-02-12-103940.img was successfully created.
pi@pi-ager:/usr/local/bin $
```

4. Zip and deploy your new image.